Antenna.

The radiation intencity $U = r^2 W_r = A_0 \cdot \sin^2\theta$

$$D_{max} = \frac{U_{max}}{U_{ave}} = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi \cdot A_0}{\frac{8}{3}\pi A_0} = \frac{3}{2}$$

$U_{max} = A_0$

$$P_{rad} = \int_0^{2\pi}\int_0^{\pi} A_0 \sin^2\theta \cdot \sin\theta \, d\theta \, d\phi = A_0 \int_0^{\pi} \sin^3\theta \, d\theta \cdot 2\pi = \frac{8}{3}\pi A_0$$

$D = D_0 \sin^2\theta = 1.5 \sin^2\theta.$